

LECTURE 6: INTRODUCTION TO GAME THEORY

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WEEK 3

AMERICAN CARS ARE GETTING BIGGER AND BIGGER

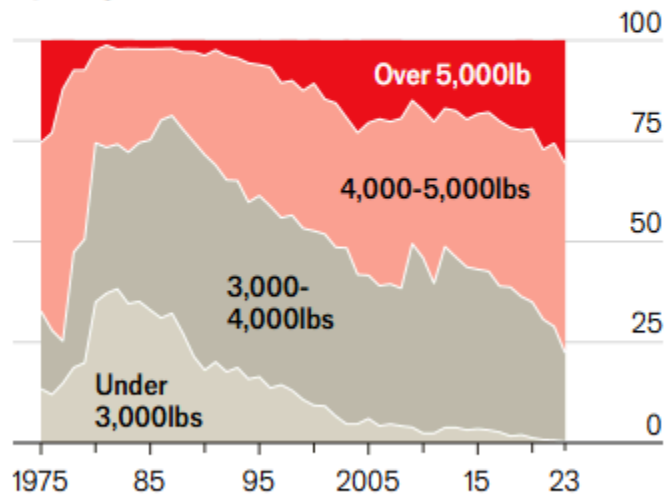
The following is a story from the Economist available [here](#)

The cars that Americans are buying are getting heavier and heavier

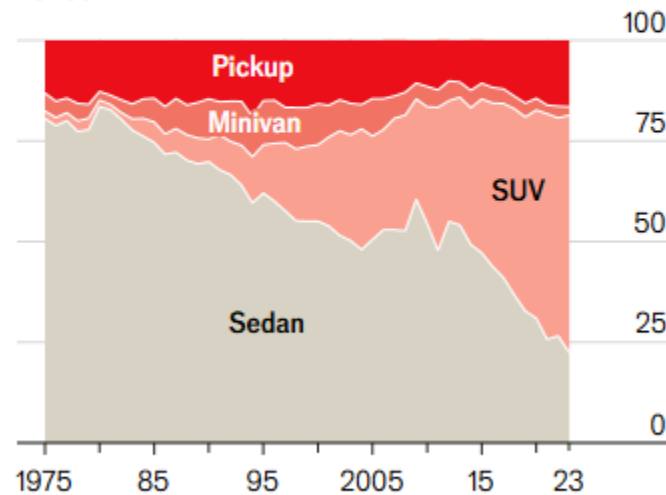
Bulk-buying

United States, new vehicle production

By weight, %



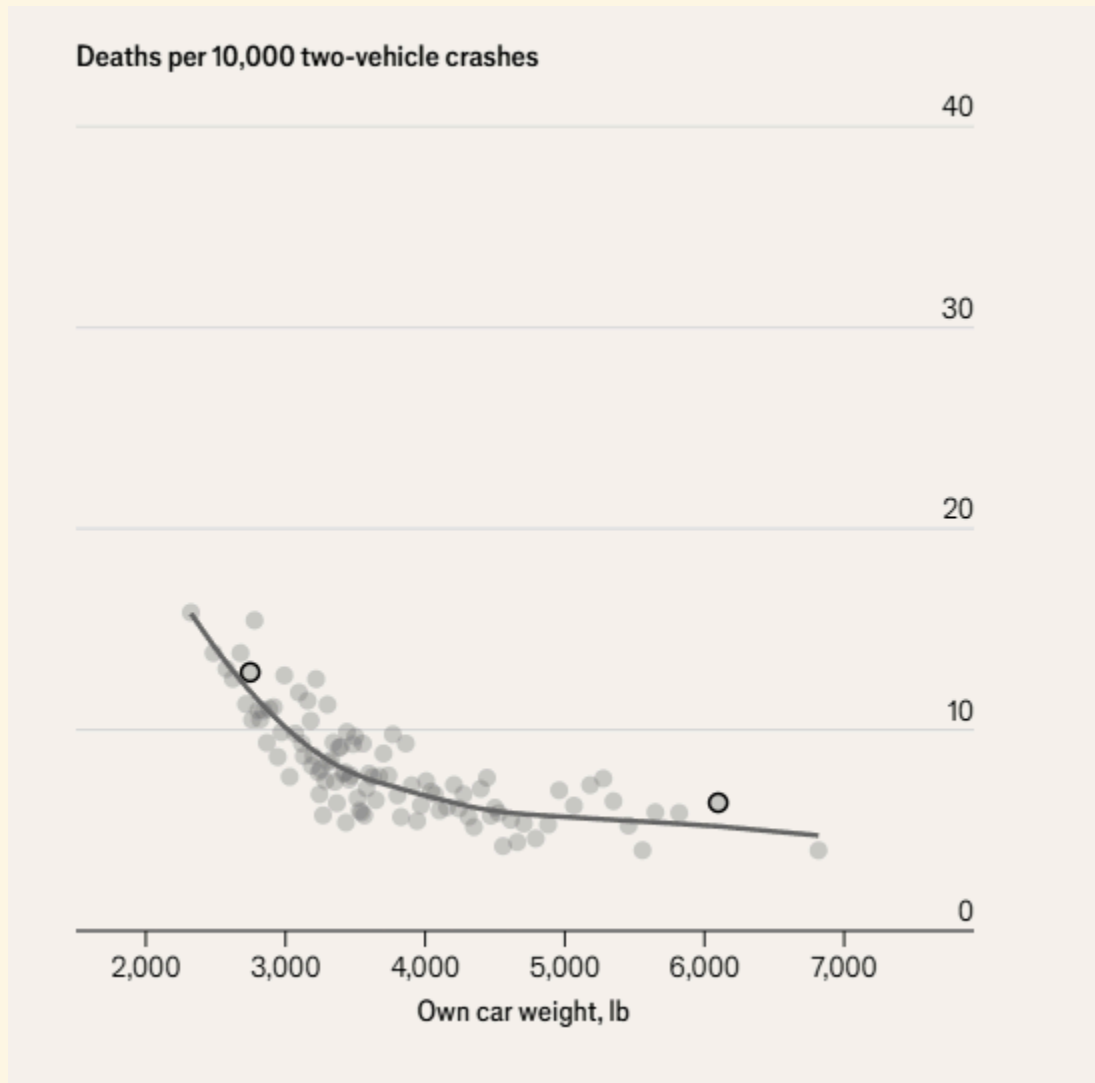
By type, %



Source: Environmental Protection Agency

THERE IS A RATIONAL THOUGHT HERE

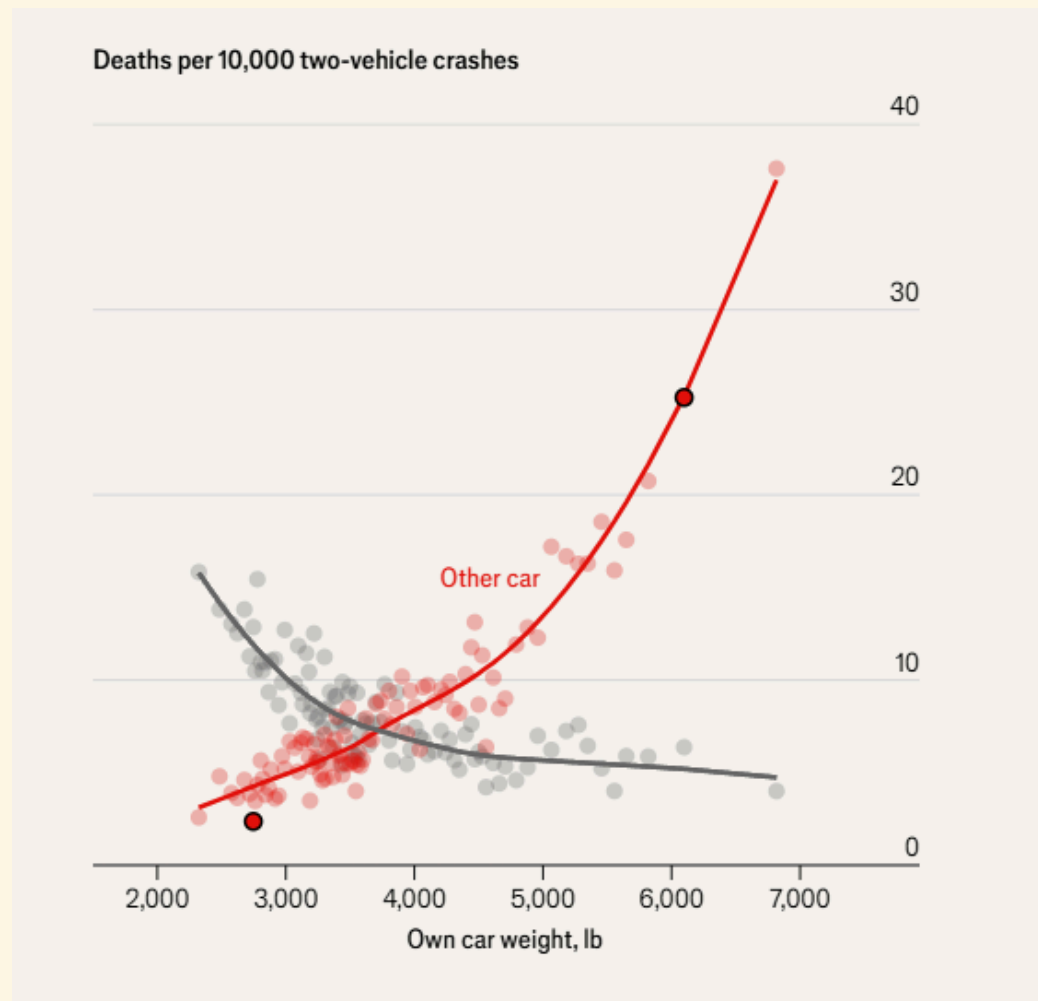
Generally speaking, if you're in a heavier car, you're less likely to die if you're in a crash



BUT THIS MAKES EVERYONE ELSE LESS SAFE

But if you're in a heavy car, you're much more likely to kill someone if you're in an accident

If you have a Ford F350, the probability that someone in the other car dies is 35%, compared to 5% if you're in a Honda Civic



LET'S FOLLOW THE TRAIL OF INCENTIVES

We have a situation where people are responding to each other's behaviours:

- If Anna and Beth have small cars, they are both quite safe
- Anna thinks they will be more safe with a bigger car, so they buy a bigger car.
- This makes Beth incredibly unsafe in their small car!
- So Beth buys a big car too!
- Now Anna and Beth both have a big car but are more unsafe than when they both had small cars!

Key point: Anna and Beth's decisions are influenced by one another's

PRISONER'S DILEMMA

This is an example of what we call a 'Prisoner's dilemma' (you'll see why in discussion)

Both Anna and Beth are less safe both having big cars, but they both got there through rational decisions

We're going to introduce some concepts in 'game theory' to explain this

CAR SAFETY AND GAME THEORY

	Anna buys a small car	Anna buys a big car
Beth buys a small car	Anna 10% probability of dying Beth 10% probability of dying	A: 5% B: 30%
Beth buys a big car car	A: 30% B: 5%	A: 15% B: 15%

The above is a **payoff matrix**. It shows the decisions of both Anna and Beth, and their 'payoffs' - their probability of dying if they're in a crash

CAR SAFETY AND GAME THEORY

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From our story, we ended up where both Anna and Beth bought big cars. But they both would have been safer with small cars. How did we end up here?

CAR SAFETY AND GAME THEORY

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What we have showed is that both buying big cars is a **Nash Equilibrium** - keeping the other player's decisions the same, no one would want to deviate from this

If Anna has a big car, Beth prefers having a big car over a small car

If Anna has a small car, Beth still prefers having a big car over a small car

NASH EQUILIBRIA IN GAMES

- We setup our games that players choose decisions at the same time, without knowing what the other person chooses
- After the game is played:
 - If anyone regrets their action, would prefer to have 'deviated', then we don't have an equilibrium
 - If no one regrets their action, we call this a **Nash Equilibrium**

NASH EQUILIBRIA IN GAMES

- We should think of finding Nash equilibria like finding the optimal point in our profit function
- There is no other action I could take that would make me better off **keeping everyone else's decisions the same**
- **Of course** this is harder than, say, finding the optimal price as a monopolist, because other people are making decisions too
- Evolving this skill of thinking about Nash equilibria is going to help us think about a small number of firms competing with each other

NASH EQUILIBRIA IN GAMES

- Every game has at least one Nash Equilibrium, some games will have multiple equilibria
- Sometimes that equilibrium will require players make decisions with some degree of randomness
- When a Nash equilibrium has no randomisation, we call it a pure strategy equilibrium
- When a Nash equilibrium requires randomisation, we call it a mixed strategy equilibrium
- The Nash Equilibria is not always the **socially optimal** outcome - we would prefer everyone to have smaller cars, but we end up with big ones

A LOT OF GAME THEORY WAS DEVELOPED POST WWII THINKING ABOUT NUCLEAR PROLIFERATION

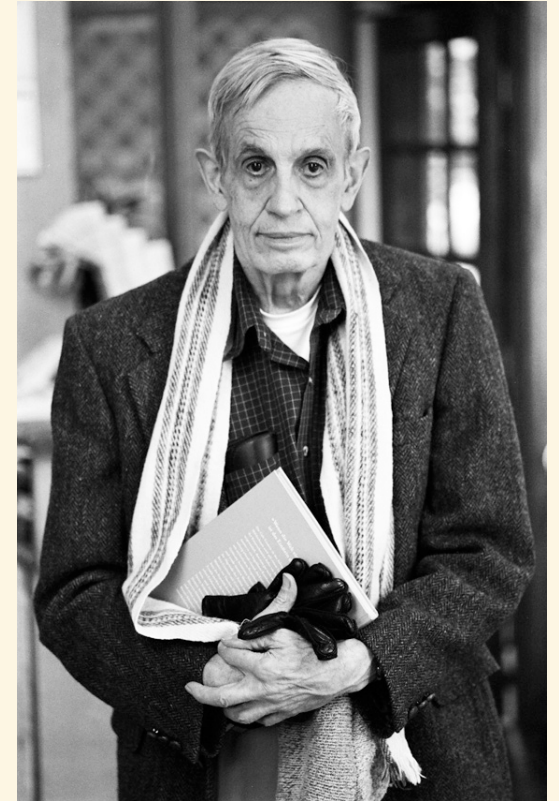
- John von Neumann was an absolute legend
- Worked as a physicist on the Manhattan Project
- Helped create the first computers. All modern computers still use the same architecture he pioneered
- Developed the ground work of game theory in the wake of WWII
 - Many people were thinking about whether nuclear weapon proliferation would lead to peace or war
 - von Neumann, among others, studied this with game theory
 - The problem was it was difficult to work out if games had natural equilibria and what they were - so would policy lead to peace or war?



ENTER JOHN NASH

- At 22, wrote a 28 page PhD dissertation showing that all games that we'll consider have equilibria (jealous)
- His math is far more complicated than anything we'll look at, but gives us the general idea:

Any game we look at, will have at least one equilibrium point



LET'S SEE HOW WE COULD SOLVE TO FIND NASH EQUILIBRIA

	Firm 1 charges a high price	Firm 1 charges a low price
Firm 2 charges a high price	F1 gets 10 F2 gets 10	F1 gets 12 F2 gets 0
Firm 2 charges a low price	F1 gets 0 F2 gets 12	F1 gets 8 F2 gets 8

THE KIT KATS ARE BACK

I need two people to play Share or Steal

- There are six kit kats on offer
- Players can choose to play share or steal
- If both play share, they split the loot 50-50
- If one shares and one steals, the stealer gets it all
- If both steal - both get nothing!

SHARE OR STEAL

Let's work through the Nash Equilibria here

	Person 1 plays Share	Person 1 plays Steal
Person 2 plays Share	P1 gets 3 P2 gets 3	P1 gets 6 P2 gets 0
Person 2 plays Steal	P1 gets 0 P2 gets 6	P1 gets 0 P2 gets 0

SOMETIMES THERE MIGHT BE MULTIPLE NASH EQUILIBRIUM

	Anna buys a small car	Anna buys a big car
Beth buys a small car	Anna 10% probability of dying Beth 10% probability of dying	A: 12% B: 30%
Beth buys a big car car	A: 30% B: 12%	A: 15% B: 15%

Slightly changing the probabilities now means that both buying small cars, or both buying big cars are an equilibrium

